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HighGen—Comparative genomics resources for cannabis

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UCSC's chain/net and MULTIZ tools are tuned for cross-species vertebrate panels. With **HighGen**, we adapted them in Galaxy across **217** ***Cannabis sativa*** **assemblies** anchored on cs10 through standardized repeat masking, a held-out test of an empirical substitution matrix, chunked extension, and streaming 64-bit dynamic programming. We pair the alignment with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

Galaxy PROJECT

220-assembly Hub

Attach genome.ucsc.edu/hubspace/74/richard%252Dburhans/highgen_cannabis/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (63 to >70%, mostly LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025, Pike et al. 2026). Projecting 216 query assemblies onto the linear cs10 (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025) seeds only through uppercase bases

2 Cross-species matrix
HOXD70 (Chiaromonte et al. 2002) was trained below 70% identity, where 49.7% of pooled *Cannabis* pairs fall (Ts/Tv = 0.872 vs. a 0.74 chance level)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse netFilter -syn (Kent et al. 2003, maxFar = 200 kb) prunes rearranged tandem gene clusters

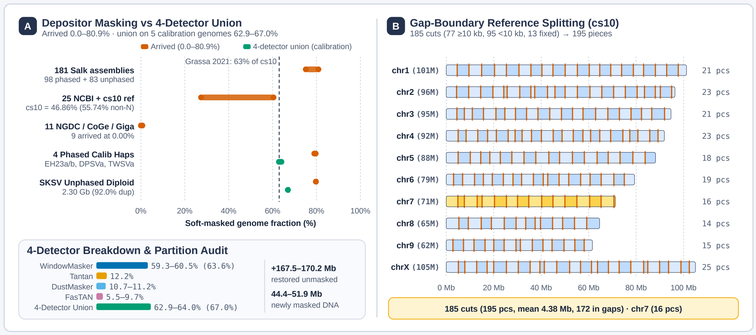


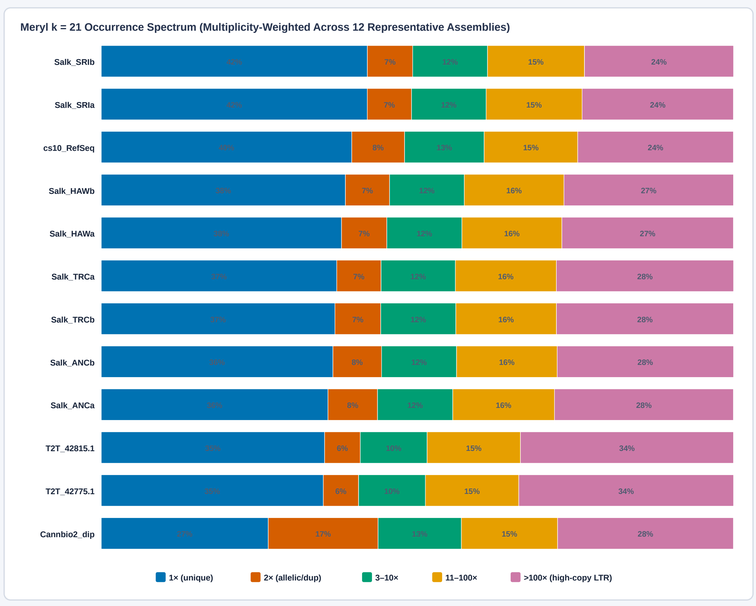
Figure 1 | Standardized masking and gap-boundary reference splitting. Panel **A**: arrived masks (0.0–80.9%, 218 assemblies) vs. the 4-detector union on 5 calibration genomes (62.9–67.0%), line at Grassa et al.'s (2021) 63% for cs10. Panel **B**: across cs10, 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts in wfi_cs10_splits.bed) partition the assembly into 195 ~5 Mb pieces plus 26 unplaced scaffolds, mitochondrion NC_029855.1, and small.00. mafSplit divides straddling blocks without base loss, and maf_concat rejoins the pieces.

B Standardized 4-detector repeat masking

We curated 221 assembly records—217 non-redundant in MULTIZ (cs10 plus 216 queries across 164 prefix groups and 156 plants, scored with rosales_odb12 BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026). To replace incompatible depositor masks, our brc-tools UDT wrappers strip arrived masks and unite four *de novo* detectors (Morgulis et al. 2006a, 2006b, Frith 2011, Myers 2026):

fasta-uppercase → WindowMasker DustMasker Tantan FasTAN → bedtools merge

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome pilot: 22.61 → 4.85 pt spread)	Depositor-only masking frees ~22% of each calibration haplotype to seed (repeat status unmeasured)
25 NCBI GenBank + cs10	27.42–60.43% (cs10 46.86% total, 55.74% non-N)		Leaves 44.26% of cs10 non-N bases open to seeding
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Unmasked queries seed across all of their bases
4 Phased Calibration Haps (EH23a/b, DPSVa, TWSVa)	**79.00–79.61%**	**62.90–64.02%** **(1.12-pt window)**	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid SKSV (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (cs10, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% 1× and 6–8% 2× multiplicity-weighted *k*-mers, whereas uncollapsed diploid Cannbio-2 (Braich et al. 2020) carries 27% 1× and 17% 2×.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two phased haplotype pairs (EH23a × EH23b and DPSVa × TWSVa, 95.3% rosales_odb12 BUSCO completeness, 1.8–2.0% duplication). We restored her CDS excision (–7.3% of chained segments) and single-chain colinearity (--max-chains 1), and inverted the training window to [85, 95)% identity, 73.9% of whose bases are cross-chromosome. The filter enriched same-chromosome over cross-chromosome matches **8.2-fold** on a Cb2 × cs10 slice (54.5% vs. a 6.6% median) and **10.9-fold** over all 200 EH23a × EH23b groups (8.08% vs. 0.74%).

331.05M
Pooled [85, 95)% pairs, symmetrized log-odds (Ts/Tv = 2.119, H = 1.4140 bits, either pair alone within 1 unit)

0/10 vs. human-chimp
UCSC's human-chimp matrix beat ours on the first 20 Mb of all ten held-out chromosomes (9/10 by minimap2). Ours beat HOXD70 10/10 by colinearity, 3/10 by minimap2

5,033×
axtChain shuffled-null suppression (EH23a chr1 × *Morus*) + scaled 682/51 gaps (70.6% single coverage)

We normalized scores to a maximum match of 100 (+80 A/T, +100 C/G, –92 transitions, –191 A-C/G-T, –197 C-G, –198 A-T transversions). Across 40 false-match budgets (colinearity referee, gap-free), the matrix beat unit ±100 on **10/10** at 20 Mb, **10/10 loci**, and **62/70 windows**. Its gain over the unchained matrix was small (0.02–0.11% at 20 Mb) and 6/10 under minimap2. Production used it for all 218 runs.

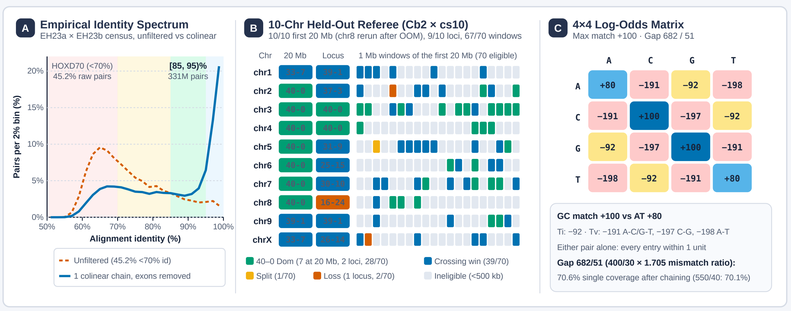


Figure 2 | Intra-species matrix calibration and 10-chromosome × 20-window held-out referee. Panel **A**: empirical identity spectra and single-chain colinearity separation. Panel **B**: colinearity-referee wins vs. HOXD70 on the first 20 Mb of 10 held-out Cb2 × cs10 chromosomes and 70 1 Mb windows. Panel **C**: symmetric 4×4 log-odds weights with 682/51 gap scaling.

D Collection-native LASTZ and MULTIZ 11.3/11.4

- **Growler LASTZ (growler_ucsc_pair, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed .2bit archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into ≤50,000-segment chunks cuts EH23a × EH23b gapped extension **7.8-fold** (389.0 → 49.8 slot-h, 1.79 h for 100 jobs, 18% above batched_lastz, median 28 min/job, under 3.97 GB RAM). growler_lastz_lav, which chunks identically, reproduced **99.99%** of UCSC's hg38 × panTro6 chr21 all.chain and rbest.chain pairs, and doRecipBest chains on one query per class.
- **Storage and 218-run production cohort:** Purging transient intermediate collections after workflow completion cut total Galaxy history storage across all 218 runs (54,554 jobs, 10,325 job-hours) from **14.96 TB to 6.50 TB (–56.5%,** 8.45 TB saved, summed). Syntenic nets cover a median **65.2%** of cs10 non-N bases across 216 queries (range 12.2–85.1%, vs. **70.0%** and 29.3–88.1% in rbest, with 40.0% diploid vs. 54.0% phased median query-side coverage).
- **Streaming 64-bit column-count MULTIZ 11.3/11.4:** In v11.3 (multiz#19, Blanchette et al. 2004), coordinate-ordered streaming cursors replaced in-memory MAF buffering (v11.2) and int64_t accumulators fixed 32-bit overflow across 29 deep blocks on NC_044376.1.02 (24 with scores ≥2^{31}). Peak memory fell **51.7-fold** on hg38 5-way chrX (677 → 13 MiB). In v11.4 (multiz#28), factoring mz_yama.c DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) sped up one upstream-timed deep merge **33.8-fold** (981 → 29 s). On one idle host, a whole 4.4 Mb piece took 3.9 → 0.80 h (**4.9-fold**, medians of three interleaved runs per release). Output is identical apart from # lines.

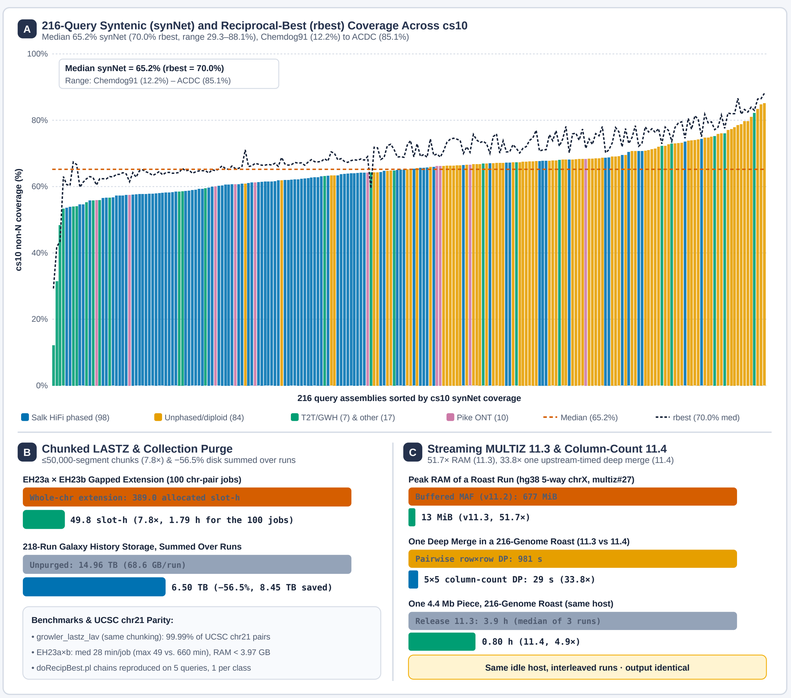


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Panel **A**: sorted synNet (median 65.2%, range 12.2–85.1%) and rbest 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class. Panel **B**: chunked LASTZ speedup, hg38 chr21 parity, and 218-run storage purge. Panel **C**: MULTIZ v11.3/v11.4 streaming RAM reduction (51.7×, human 5-way) and column-count DP speedup (33.8×, one merge).

E Tandem metabolic clusters and syntenic net limits

- **cs10 tandem census and permutation audit:** Under MCScanX tandem criteria (Wang et al. 2012) adapted to stricter BLASTp thresholds ($E \le 10^{-10}$, top 10 hits), **3,035** of cs10's 25,296 RefSeq protein-coding genes reside in 1,208 tandem arrays (K = 0) and **5,123 (20.3%)** in 1,796 clusters ($K \le 10$). These clusters include **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including CBDAS). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs. 11.0% controls, p = 0.022) does not survive family-wise correction (p_{FWER} = 0.12, threshold-free p = 0.40, BBE-like 2.8%, p = 0.59).
- **Functional CBDAS vs. B-locus pseudogenes:** At cs10 functional CBDAS (LOC115697762, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length CBDAS) at ≥98.5% identity and **6 assemblies** below 93.0% (≥50% CDS) from **77 assemblies** (67 plants) at 93.0–98.5%. In that band, 63 carry Lynch et al. (2025) annotations placing the alignment on the ~94% B-locus pseudogene that Laverty et al. (2019) described (62 in THC-dominant cultivars), and 14 carry none.
- **netFilter -syn pruning vs. paralog reuse:** Applied beyond UCSC's contiguity rule, its human-mouse cutoffs (maxFar = 200 kb, plus minTopScore = 300,000 for 14.5% of losses) pruned **4.7% of TPS** (418/8,856: 4.2% clustered, 6.4% singleton) and **10.7% of BBE-like** (807/7,560: 7.5% clustered, 26.2% singleton vs. 5.2%/13.1% controls). In synNet, this dropped 54 CBDAS assemblies (under 50% CDS), including Finola's intact 99.4% transposed copy (CM011610.1:21.84 Mb, 26.8 Mb from its enclosing gap). Removing netFilter -syn leaves **46.4%** of recovered pairs (569/1,225) overlapping another family gene's interval. Resolving them needs rbest nets or recalibrated thresholds with duplication-aware orthology (TOAST/BOAST, Hou et al. 2006, 2009).

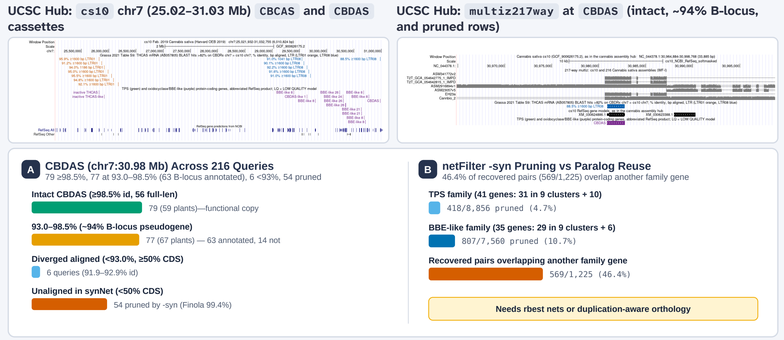


Figure 4 | Live HighGen UCSC Track Hub on chr7 and netFilter -syn trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and multiz217way at CBDAS (right). Below: (**A**) 216-query CBDAS stratification and (**B**) netFilter -syn pruning vs. the 46.4% of recovered pairs overlapping another family gene.

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (panagram_index / panagram_view, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: cs10 + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected k = 21 over k = 31 on higher 1 Mb window Pearson r with rbest identity across **16/16 Salk haplotypes** (mean r = 0.7150 vs. 0.6901, Δr = +0.0249 within the 0.05 equivalence band). Tier P2 indexed **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index). It passed $r \ge 0.50$ in **41 of 42 chain-scored queries** (r = 0.5848–0.8118, Pink pepper r = 0.7376) and separated non-PAR chrY cleanly (3/3 males 0.976–0.987 vs. 3/3 V3 females ≤ 0.0773 across 205,619 subsampled AH3Ma-absent positions). It flagged 2 Guppy 5 SUP trio-binned ONT haplotypes on G3 count tolerance (Punto Rojo –10.4%, Cherry Pie –10.8%, Pike et al. 2025). Only one Sri Lankan ONT haplotype fell below 0.50 (ASM5191173v1, r = 0.4763, Merqury QV 24.01/24.11, Pike et al. 2026, for reasons still under investigation.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On cs10, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion NC_029855.1) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried all.chain datasets (1.45 TB). On its zero-divergence identity-chain control, TOGA2 called **25 of 24,503 lost/uncertain (0.10%)**—1 L + 24 UL against a ≤ 0.5% bar. On EH23a, the production, loose, and RepeatFiller-filled-and-cleaned chains called **2,077 (8.48%)**, **1,778 (7.26%)**, and **1,660 (6.77%)** standard genes L/UL. Filled and production arms disagree on 713 of 2,161 L/UL/M genes (33.0%), while production yields the fewest multi-copy false presences (4,306 standard genes vs. 5,058 loose and 5,245 filled). Because our top-ranked native-model check (B6) is still pending, the pilot selects no arm and reports no gene loss.
- **Pedigree-verified phenotype ledger:** We joined 40 sources across 215 assemblies—30 publications (29 papers, including 2 supplements, plus 988 Lynch et al. 2025 Figshare rows across 190 assemblies from 43 audited extractions in passed.tsv) and 10 NCBI BioSamples. This yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons). Under strict measured / reported filtering, 0 of 14 traits pass our ≥30-family, ≥2-class (≥5 families/class) association screen. Varin spans 30 families (37/38 non-WT plants in HO40 across 3 classes), while Chemotype reaches 30 families with 2 classes ≥5 only if Lynch's HPLC methods sentence covers its table column.

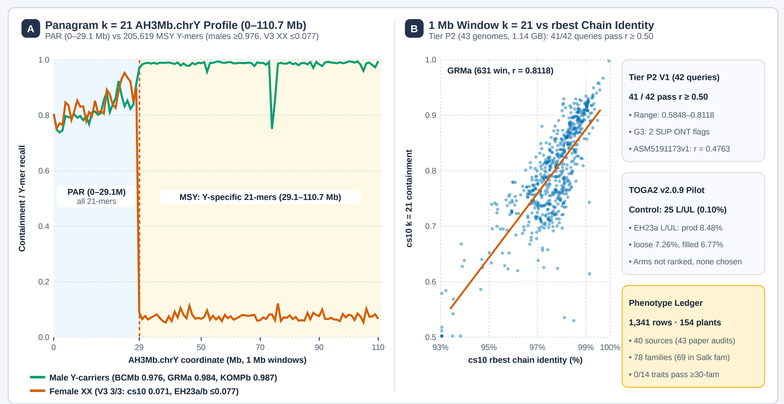


Figure 5 | Alignment-free Panagram chrY profile and rbest correlation. Panel **A**: chromosome-wide AH3Mb.chrY (0–110.7 Mb) k = 21 profile of PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions. Panel **B**: all 631 eligible 1 Mb windows on GRMa (r = 0.8118) with Tier P2, TOGA2, and phenotype scorecards.

Public tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → brc-tools GPU seeding + chunked LASTZ → growler_lastz_lav 217-way streaming TBA → MULTIZ 11.3/11.4 Pan-*k*-mer + orthology → Panagram and TOGA2

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HighGen Hub | Galaxy Tools | Repository